

Primer on 802.11n

802.11n, an amendment of the 802.11 standard, promises significantly improved throughput and range as compared to the existing 802.11 standards. Most devices tested in this comparison are based on a stable draft version called as the 802.11n draft 2.0. Earlier versions are called 'Pre-n' and they were plagued with interoperability issues.

802.11n includes many new features that improve the throughput, range and reliability of wireless networks. The adjoining table gives an overview of the core technical difference between the various 802.11 standards.

From the above table, as is clearly evident, 802.11n devices can work on both 2.4- as well as 5-GHz frequency bands, and hence, anyone using them has an option of using it on either frequency range. When used at 5 GHz, the network is less susceptible to interference because very few electronic devices transmit in that frequency range. Also, the 5-GHz band offers 23 channels as compared to 3 channels offered by the 2.4-GHz band. Hence, it is easy to find an interference-free channel. Moreover, the 5-GHz band can support eleven 40-MHz channels, thus boosting the aggregate bandwidth available to the device. The same device, when used in the 2.4-GHz band, can only support one 40-MHz channel. A point worth noting is that the WiFi alliance doesn't certify the use of 40 MHz in the 2.4-GHz band.

Other important features that 802.11n offers is Frame aggregation, MIMO and channel bonding.

Frame Aggregation: 802.11n improves MAC layer performance by allowing devices to aggregate several packets into one single packet using two different techniques. The wasted overhead between frames is thus avoided due to this aggregation. Hence, 802.11n devices require less air time as compared to devices based on 802.11 a / b / g standards.

MIMO: Stands for Multiple Input, Multiple output, and is a technology

	802.11a	802.11b	802.11g	802.11n (Draft 2.0)
Maximum throughput	54 Mbps	11 Mbps	54 Mbps	300 Mbps
Operating frequency	5 GHz	2.4 GHz	2.4 GHz	2.4 and 5 GHz
Non-overlapping channels	23	3	3	3 at 2.4 GHz, 23 at 5 GHz
Standard Approved	Yes	Yes	Yes	Expected by 2009
WiFi certified	Yes	Yes	Yes	Yes (Draft 2.0 devices)
Interference Sources	Cordless phones using 5 GHz channels	Bluetooth, microwave ovens etc	Bluetooth, microwave ovens etc	Same as 802.11a at 5 GHz and same as 802.11 bg at 2.4 GHz

where multiple antennas are used at both the transmitter as well as the receiving end. It has the advantage of radio frequency signal property called 'Multipath', that is, RF signals often reflects off objects in their path, thus creating multiple streams of the same signal, but separated by time in the spatial domain. MIMO uses a technique called Spatial multiplexing, where the signal is transmitted in separate data streams at the same frequency, but over different spatial channels. Hence, 802.11n devices perform excellently in an environment that causes lot of multipath reflection.

Channel Bonding: Existing standards such as 802.11a and 802.11g use a single 20 MHz channel, whereas 802.11n can bond two such 20 MHz channels to form a single, wide band 40 MHz channel, thus improving effective bandwidth and boosting the maximum throughput.

Operational Modes

Since 802.11n is backward compatible with the existing standards, supporting devices can be made to run in three different modes,

Mixed Mode: This mode enables 802.11n devices to co-exist and interoperate with legacy 802.11 a / b / g devices on the same wireless network. This may not be important to a home user, but in a small office, which already has legacy WiFi hardware (802.11 a / b / g) devices this mode ensures interoperability.

Legacy Mode: Here the 802.11n-based router / access point will operate in 802.11 a / g mode, but with some improved performance due to the

enhancement done at the physical layer as defined in the 802.11n Draft 2.0. This mode is important if you buy a new 802.11n router, but have laptops with legacy 802.11 a / b / g wireless cards and don't want to enable 802.11n in your router.

802.11n Mode: Here, the 802.11n router will work as a pure 802.11n device and won't accept connections from 802.11 a / g devices. This kind of operation can be used for best throughput for laptops based on 802.11n wireless cards.

Agent001's tips for 802.11n WiFi network:

1. Before buying any router / Access point, ensure it has minimum two radios—dual radio or 2x2 matrix, 3x3 matrix is still better.
2. To ensure good flexibility, buy dual-band devices, that is, devices which support for both, the 2.4- as well as 5-GHz frequency band.
3. Whenever possible, configure the 802.11n device to use the 5-GHz band. This will improve the maximum throughput, albeit with decreased signal coverage as compared to 802.11n at 2.4 GHz. Channel bonding will help and hence select the 40 MHz settings when configuring the device.
4. Avoid keeping the router near known interference devices (Cordless phones, microwave ovens, etc) working at same frequency band.
5. If you already have too many laptops with 802.11 a / g hardware, configure your new 802.11n router to run in mixed mode for better compatibility.